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# Entry Without Displacement: How Open-Weight Models Compete in the Market for Inference

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## Abstract

1 Open-weight models are widely expected to constrain the pricing power of propri-  
2 etary model providers, but the mechanism has not been measured. We assemble a  
3 market-wide panel of the models served by a large inference router: posted prices,  
4 capability benchmarks, weight-release status and revealed usage for 457 models,  
5 extended back to 2023 with archived catalogues. Three facts emerge. Competition  
6 works through releases rather than repricing: 54% of models are never repriced  
7 at all, and a posted price changes in only 12.6% of model-months, even though  
8 changing it costs a provider nothing. When prices do change, volume responds  
9 weakly, with an own-price elasticity of  $-0.53$  (s.e. 0.10) identified from 708 re-  
10 visions with model and date fixed effects. Entry does not displace incumbents:  
11 over four weeks, models released inside the window took 28% of tokens while  
12 incumbent volume grew 18%, and the confidence interval rules out differential  
13 losses above 0.13 log points for the incumbents closest to an entrant in capability  
14 space. Removing open-weight models from the choice set in an estimated nested-  
15 logit demand system raises the average price paid per token by 156% and costs  
16 users about \$2.4m per day on this platform, while raising provider concentration.  
17 Open weights discipline the price at which new demand is served, not the prices  
18 incumbents post.

## 19 1 Introduction

20 Whether open-weight models constrain the market power of proprietary model providers is now a  
21 policy question as much as an economic one. Reports commissioned by governments name it as an  
22 open evidence gap [Bengio et al., 2026], and the theoretical case cuts both ways: scale economies in  
23 training push toward concentration [Korinek and Vipra, 2025], while freely downloadable weights  
24 supply a competitive fringe that any hosting provider can serve. Recent empirical work has begun  
25 to describe the market. Demiret et al. [2025] document a collapse in the price of a unit of measured  
26 intelligence and a proliferation of suppliers; Nagle and Yue [2025] show that a large share of propri-  
27 etary usage is directed at models that are dominated on price and measured capability by an open  
28 alternative, and estimate the savings foregone.

29 Both of those results are about the allocation of demand across models at a point in time. Neither  
30 identifies the channel through which open weights would constrain proprietary pricing, and the  
31 two natural channels have opposite implications for policy. If open weights force incumbents to  
32 cut posted prices, then the constraint is visible in the price series of established models and its  
33 strength can be read off their price responses. If instead open weights capture new demand while  
34 incumbents hold their prices, the constraint operates on the terms at which the market grows, and  
35 any measurement based on incumbent price responses will find nothing.

36 We assemble a panel that can separate the two. Our data come from a large inference router that  
37 publishes model-level usage, prices and capability scores. We combine the current cross-section  
38 (457 models with observed usage, 175 of them open-weight), a daily usage panel covering all of  
39 them for one month, task-level market shares for 29 distinct uses, per-provider hosting terms, and a  
40 panel of archived price catalogues and usage leaderboards reaching back to 2023. The panel used  
41 for price responses contains 19,803 model-date observations on 634 models.

42 Three facts follow, and they point the same way. First, the posted price of a given model version  
43 behaves like a fixed characteristic of the product: 54% of models are never repriced, and the monthly  
44 frequency of a price change is 12.6%, comparable to consumer goods [Nakamura and Steinsson,  
45 2008, Cavallo, 2018] in a setting where the menu cost that explains rigidity there does not exist.  
46 Second, when prices do change, volume moves little. Exploiting 708 revisions with model and date  
47 fixed effects, we estimate an own-price elasticity of  $-0.53$  (s.e. 0.10), below the value near one  
48 reported by Demirer et al. [2025] from cross-model variation and far below what homogeneous-  
49 goods reasoning would imply. Third, entry is absorbed by growth rather than by displacement. Over  
50 the four weeks of the daily panel the market grew 65%; models released inside the window took 28%  
51 of tokens, of which 5 of 10 releases were open-weight, and yet incumbent volume rose 18%. In an  
52 incumbent-by-entrant event study we find no differential decline for the incumbents technologically  
53 closest to an entrant, with a confidence interval that excludes declines larger than 0.13 log points.

54 Read together, these facts say that the competitive constraint from open weights does not show up  
55 where one would look for it. We therefore quantify it where it does show up. We estimate a nested-  
56 logit demand system over the 172 models with prices and capability scores, invert it to recover mean  
57 utilities that reproduce observed shares, and delete open-weight models from the choice set. Users  
58 would pay 156% more per token, and the surplus loss is worth about \$2.4m per day on this platform  
59 alone. The loss is not evenly spread: across 29 task-level markets, the implied price increase rises  
60 almost one for one with the open-weight share of task spending, from 18% on average to more  
61 than twice that in agentic and coding work. Concentration moves the other way from the usual  
62 presumption: removing open weights raises the provider Herfindahl index from 897 to 1048.

63 Our contribution is to identify the margin on which open-weight competition operates and to price it.  
64 The measurement of price rigidity and of the within-model demand response is, to our knowledge,  
65 new to this market, and it explains why the reallocation gains documented by Nagle and Yue [2025]  
66 persist: with inelastic short-run demand, a provider has little to gain from cutting the price of a model  
67 already in the field, and users have little reason to move quickly. We also release the collection code,  
68 which reconstructs the entire panel from public endpoints.

## 69 2 Data

70 **Sources.** We use the public API and web endpoints of OpenRouter, a router that serves requests  
71 to hundreds of models from many providers through one interface. Four endpoints supply the raw  
72 material: a catalogue giving posted input and output prices, context length, release timestamp and  
73 the Hugging Face repository for models whose weights are published; a leaderboard giving tokens,  
74 requests and tool calls by model; a benchmark table reporting third-party capability scores; and  
75 a task table giving each model’s share of spending in 29 usage categories over a 30-day window.  
76 Each model’s own page embeds a daily usage series for the trailing month, which we collect for  
77 every model with observed usage. We take per-provider hosting terms from the endpoint listing, and  
78 parameter counts from Hugging Face. Historical coverage comes from Internet Archive captures of  
79 the catalogue and of the leaderboard; the latter embed the leaderboard payload in the page’s streamed  
80 data, which we parse directly. Appendix A documents each step.

81 **Openness.** We classify a model as open-weight if the catalogue records a public weight repository  
82 for it. The classification is model-specific rather than provider-specific, which matters here: several  
83 providers, including Google, OpenAI and Alibaba, ship both open-weight and proprietary models,  
84 and a provider-level rule would misclassify them.

85 **Capability.** Capability enters as a vector of third-party scores covering reasoning, coding and  
86 agentic use, supplemented by pairwise-comparison scores from a design-oriented arena. These mea-  
87 sures are highly collinear. The first principal component of the three benchmark dimensions explains  
88 98% of their variance, and 91% when the two arena blocks are added, so we summarise capability

**Table 1:** The estimation sample, measured on the last complete day of the daily panel. Prices are dollars per million tokens for the blended input-output mix. Capability is the first principal component of the benchmark block. Shares are within the estimation sample, which covers 91% of platform tokens.

|             | Models | Median price | Median capability | Median context (thousands) | Token share | Spend share |
|-------------|--------|--------------|-------------------|----------------------------|-------------|-------------|
| Open-weight | 86     | 0.29         | 30.5              | 256                        | 0.73        | 0.32        |
| Proprietary | 86     | 1.50         | 36.6              | 400                        | 0.27        | 0.68        |
| All scored  | 172    | 0.59         | 33.4              | 262                        | 1.00        | 1.00        |

by that first component and report the underlying dimensions in the appendix. Scores are missing for smaller models; we impute a missing dimension by ridge regression on the dimensions a model does have, which recovers held-out values with  $R^2$  above 0.96. The resulting sample of 180 scored models accounts for 91% of tokens.

**Prices.** Providers post separate input and output prices. Because the input-output mix differs sharply across models, we work with a blended price that weights the two by the observed token mix, and with a fixed-weight index that holds the mix at its sample mean when we need a price series free of feedback from usage. Table 1 summarises the estimation sample.

**What the market is.** The platform is one segment of the market for inference. It excludes traffic sent directly to a provider’s own API and consumer subscription products, and its users are developers who have chosen a router, which selects for price sensitivity and for willingness to switch. Our estimates therefore describe the segment in which open-weight substitution is easiest. If anything this biases us toward finding a strong competitive constraint, which sharpens the interpretation of the weak responses we report below.

### 3 Three facts

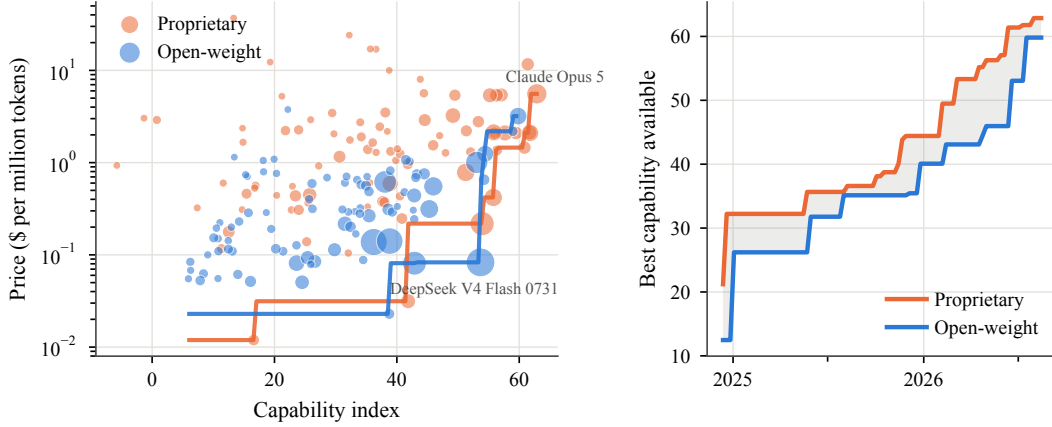
#### 3.1 Capability is cheap, and open weights are cheaper

Figure 1 places every model with usage on the price-capability plane. Two features stand out. At any capability level below the frontier, prices differ by one to two orders of magnitude, so posted price is far from a sufficient statistic for what users buy. Second, the open-weight side of the market is systematically cheaper. Matching each proprietary model to the open-weight models within two points of it on the capability index, the median proprietary model costs 3.3 times the median open alternative, or 1.7 times when the comparison is weighted by volume. Averaged over actual usage, a token from an open-weight model costs \$0.35 per million against \$1.95 for a proprietary one. These gaps are smaller than the ninety percent difference reported by Demirer et al. [2025] and Nagle and Yue [2025] for 2025 data, which is what one would expect if proprietary providers have since released cheaper tiers; the frontier comparison drawn in Figure 1, which uses only the cheapest option available at each capability level, narrows the gap further to 1.4 times.

The right panel tracks the best capability obtainable from each side over time. The open-weight frontier follows the proprietary frontier with a median lag of 105 days. The gap is a lag, not a level: at the very top of the capability distribution proprietary models are unmatched at every date, but the set of tasks for which that matters shrinks as the open frontier advances.

The open-weight share of volume is also far from stable. In the archived leaderboards, which are internally consistent over time though not directly comparable in level with the daily panel, open-weight models held 86% of tokens in late 2023, fell to 29% by mid-2025, and recovered to 48% by mid-2026. The trough coincides with the sample used by Nagle and Yue [2025], whose finding that proprietary models dominate usage describes a market that has since moved.

Openness also changes the supply side. Weighting by volume, an open-weight model is served by 14.9 independent hosting providers against 3.0 for a proprietary model, and the price a user pays for an open model is set by hosts competing to serve the same weights rather than by the organisation that trained it. Open-weight models account for 68% of tokens but only 30% of spending, and the



**Figure 1: The price of capability.** Left: posted blended price against the capability index, with marker area proportional to the square root of token volume; the step functions give the cheapest price available at or above each capability level for each side of the market. Right: the highest capability available from each side by release date, with the shaded region marking the open-weight lag.

129 difference shows up in concentration: the provider Herfindahl index is 1183 measured on tokens and  
 130 2439 measured on spending. By volume the market looks competitive; by revenue it does not.

### 131 3.2 Posted prices rarely move

132 The archived catalogues let us follow the posted price of a model version from its first appearance.  
 133 Between consecutive captures, a median of three days apart, the price changes 2.9% of the time.  
 134 Over a model’s observed life, 54% of models never change price at all. The left panel of Figure 2  
 135 shows the survival of the launch price: 56% of models are still at their launch price after six months  
 136 and 45% after a year. Conditional on a change, the median revision is a cut of about ten log points,  
 137 and 58% of changes are cuts.

138 Expressed the way the price-setting literature does, 12.6% of model-months contain a price change,  
 139 an expected duration of about 7.4 months. That is at the low end of the range reported for consumer  
 140 goods and comparable to online retail [Bils and Klenow, 2004, Nakamura and Steinsson, 2008,  
 141 Cavallo, 2018], which is itself the surprise: the menu cost that explains rigidity in those settings, the  
 142 physical and managerial cost of changing a posted price, is close to zero when the price is a field in an  
 143 API catalogue. The distribution is also more polarised than in retail, where most goods are repriced  
 144 eventually. Here more than half of models are never repriced at all. What is missing is not the  
 145 ability to reprice but the incentive, which the next fact supplies. The consequence for measurement  
 146 is immediate: over the few weeks that a release event study typically spans, an incumbent’s posted  
 147 price is unchanged with probability near 0.9, so a design that looks for the competitive effect of open  
 148 weights in incumbent price series is looking at a variable that rarely moves.

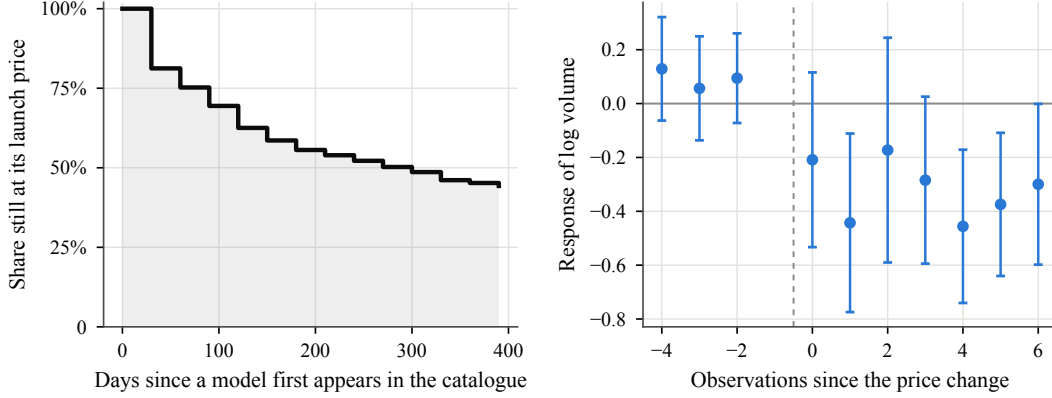
### 149 3.3 Demand for a given model is price-inelastic in the short run

150 We use the rare revisions to estimate how volume responds. Merging the archived leaderboards with  
 151 the archived catalogues gives 19,803 model-date observations on 634 models between November  
 152 2023 and May 2026, containing 708 observations that follow a price change. We regress log volume  
 153 on the log fixed-weight price with model and date fixed effects, so identification comes only from  
 154 within-model revisions, net of anything common to the market on a date:

$$\log y_{jt} = \beta \log p_{jt} + \alpha_j + \gamma_t + \varepsilon_{jt}. \quad (1)$$

155 Table 2 reports  $\hat{\beta} = -0.53$  (s.e. 0.10) for tokens and  $-0.46$  (s.e. 0.09) for requests. Open-weight  
 156 and proprietary models give indistinguishable estimates. Dropping the model fixed effects, so that  
 157 cross-model variation identifies  $\beta$ , moves the estimate to  $-0.24$ , which is what one would expect if  
 158 higher-priced models carry unobserved quality.

159 The right panel of Figure 2 traces the dynamics around 306 revisions, restricted to the captures that  
 160 report non-overlapping weekly buckets so that the path is not an artefact of a trailing aggregation



**Figure 2: Posted prices rarely move, and volume responds weakly when they do.** Left: Kaplan-Meier survival of a model’s launch price in the archived catalogues. Right: coefficients from a distributed-lag version of (1) around 306 price revisions, scaled by the size of the revision, with model-by-event and date fixed effects and 95% intervals clustered by model.

**Table 2: Own-price elasticity of demand.** Each column is a separate regression of (1). Standard errors clustered by model in parentheses.

|              | Model and date FE |                   | By openness       |                   | Date FE only      |
|--------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|              | Tokens            | Requests          | Open              | Proprietary       | Tokens            |
| log price    | −0.534<br>(0.102) | −0.463<br>(0.087) | −0.582<br>(0.116) | −0.534<br>(0.204) | −0.239<br>(0.077) |
| Observations | 19,803            | 19,803            | 8,873             | 10,930            | 19,803            |
| Models       | 634               | 634               | 321               | 316               | 634               |

161 window. Volume is flat before the change and settles about 0.3 to 0.45 log points lower per log  
162 point of price increase within a few weeks. The pre-period coefficients are small and individually  
163 insignificant, which is the relevant check: if providers raised prices after volume rose, the estimate  
164 would be biased toward zero, and we find no such pattern.

165 An elasticity near one-half is not consistent with a static Bertrand equilibrium in posted prices, which  
166 would require demand to be elastic at the optimum. We read it as a short-run object. Switching a  
167 production workload to another model requires re-testing prompts, harnesses and evaluation suites,  
168 so the relevant margin adjusts over months rather than days. It also rationalises the rigidity: a  
169 provider that would gain little volume from a price cut, and lose little from holding steady, has weak  
170 incentive to reprice a model already in the field.

## 171 4 Entry without displacement

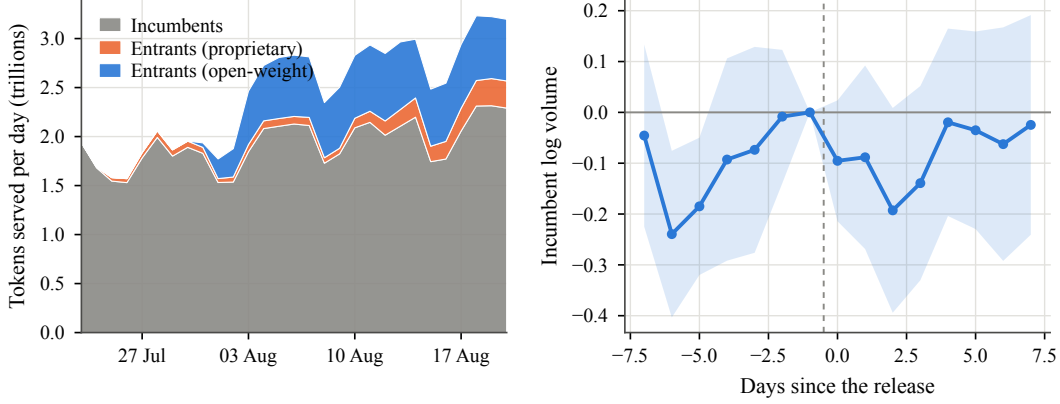
172 The daily panel covers every model with usage for the four weeks to 20 August 2026, a window  
173 containing 10 releases by providers with priced, scored models, of which 5 are open-weight. The  
174 window is unusual in one respect that turns out to be the point: the market grew 65% over it.

175 The left panel of Figure 3 decomposes daily volume into incumbents and models released during  
176 the window. Entrants reached 28% of platform tokens within four weeks, and 69% of that volume  
177 went to open-weight entrants. Incumbent volume did not fall. It grew 18%. Entry accounts for 72%  
178 of net growth, and the residual is incumbent expansion.

179 Aggregate volume can hide reallocation, so we test for it directly. For each release  $k$  at date  $T_k$  we  
180 stack all incumbents  $j$  that were already serving traffic, and estimate

$$\log y_{jt} = \beta \mathbf{1}\{t \geq T_k\} \times \tau_{jk} + \delta_{jk} + \lambda_{kt} + X'_{jt}\theta + \varepsilon_{jkt}, \quad (2)$$

181 where  $\delta_{jk}$  is an incumbent-by-entrant effect and  $\lambda_{kt}$  an entrant-by-day effect, so that  $\beta$  compares  
182 incumbents exposed to the same release on the same day. The exposure measure  $\tau_{jk} = S_{jk} r_{jk}$



**Figure 3: Entry is absorbed by growth.** Left: daily platform volume split between models already serving traffic at the start of the window and models released during it. Right: event-time coefficients from (2) on log incumbent volume, per unit of exposure to the entrant, with 95% intervals clustered by incumbent.

combines technological proximity with the entrant’s relative value for money:  $S_{jk} = \exp(-\kappa d_{jk})$  with  $d_{jk}$  the Mahalanobis distance between capability vectors, and  $r_{jk} = \log(q_k/p_k) - \log(q_j/p_j)$ . Because capability and price both trend, we also control for post-release interactions with the incumbent’s own capability and price.

We find no diversion. The estimate of  $\beta$  is 0.01 (s.e. 0.07), and the event-time path in the right panel of Figure 3 is flat on both sides of the release. The confidence interval excludes differential declines larger than 0.13 log points per unit of exposure, and tightens to 0.10 and 0.04 under the Euclidean and cosine distance measures reported in Table 4. A design with ten releases over four weeks cannot detect small effects, and we do not claim that diversion is zero. Nor can this design rule out that exposure is measured with error: if capability similarity is a poor proxy for substitutability, the null is uninformative about diversion even though it is informative about the aggregate pattern. What we can say is that in a market growing at this rate, the displacement channel is small relative to the expansion channel, which is the comparison that matters for how the constraint operates.

**A longer horizon.** Four weeks is short, and displacement plausibly takes quarters. The archived panel lets us look further out. We take the 19 open-weight releases in it that reached at least one percent of platform tokens, and compare aggregate volume of models already in the market over the eight weekly observations before and after each. Incumbent volume fell in 32% of these episodes; the median change was +0.25 log points against a median market change of +0.52. The release of DeepSeek R1 is the sharpest case: platform volume rose 0.56 log points over the sixteen weeks around it while incumbent volume was flat at −0.03. Incumbents lose share, because the market grows around them, but for the most part they do not lose volume.

## 5 What open weights are worth

Because posted prices do not respond and incumbents are not displaced, the value of open-weight competition has to be measured on the choice set rather than on prices. We estimate a nested-logit demand system over the 172 models with prices and capability scores, nesting models by capability tercile so that models of similar capability are closer substitutes than models drawn from different tiers. Writing  $g(j)$  for the nest of model  $j$  and  $\sigma \in [0, 1)$  for the within-nest correlation [Cardell, 1997, Berry, 1994], shares take the form

$$s_j = \frac{\exp(\delta_j/(1-\sigma))}{D_{g(j)}} \cdot \frac{D_{g(j)}^{1-\sigma}}{1 + \sum_h D_h^{1-\sigma}}, \quad D_g = \sum_{k \in g} \exp(\delta_k/(1-\sigma)), \quad (3)$$

with mean utility  $\delta_j = \beta_p \log p_j + \alpha' x_j + \xi_j$ , where  $x_j$  collects the capability index, context length, modality, reasoning support, latency and age. The outside option is the fringe of models we cannot score, which holds 2.4% of tokens.

Two features of (3) make the exercise tractable. Observed shares invert to mean utilities,  $\delta_j = \log s_j - \log s_0 - \sigma \log s_{j|g(j)}$  [Berry, 1994], so the system reproduces the data exactly and no part of the counterfactual depends on how well the characteristics fit. And the counterfactual depends only on  $(\beta_p, \sigma)$ : deleting the open-weight models from the choice set changes the denominators in (3), redistributing their volume across the remaining models and the outside option.

**Choosing the parameters.** We do not attempt to identify  $\beta_p$  and  $\sigma$  from cross-sectional share variation, because the instruments this literature relies on fail in this market. The differentiation instruments of Gandhi and Houde [2019] are functions of the distribution of rival characteristics, which here is a distribution over an essentially one-dimensional capability index, so they are near-collinear with a model’s own capability and therefore with its unobserved quality. The obvious cost shifter, parameter count, is available only for open-weight models and predicts unmeasured quality about as strongly as it predicts serving cost: in our data it correlates 0.65 with the capability index, and instrumenting with it moves the price coefficient from  $-0.79$  under ordinary least squares to  $-0.27$  despite a first-stage  $F$  above 900, which is the signature of a violated exclusion restriction rather than of a corrected bias. We therefore fix  $\beta_p = -1$ , between the within-model estimate of Section 3.3 and the value near one reported by Demirer et al. [2025], and report every result across a range of  $\sigma$ .

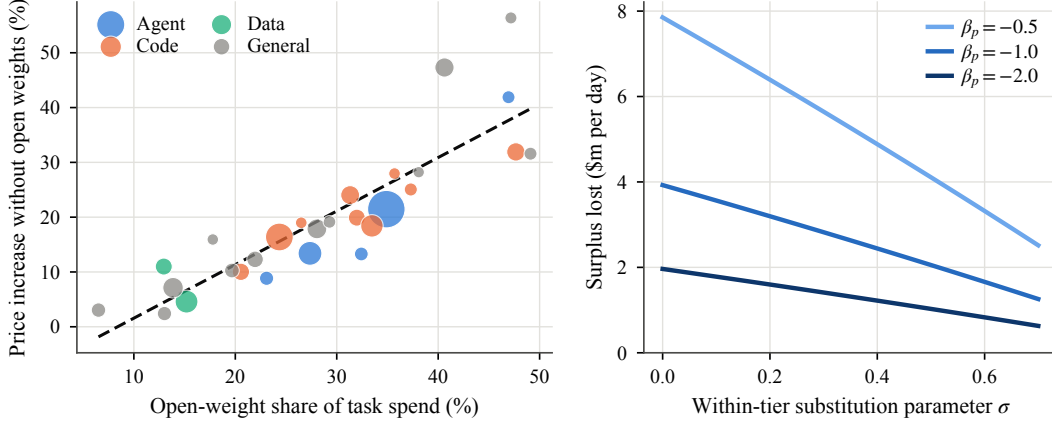
**Results.** Removing open-weight models from the choice set at posted prices, the average price paid per token rises from \$0.70 to \$1.80, an increase of 156%. Consumer surplus, computed as the log-sum expected utility of Small and Rosen [1981], falls 21%, worth about \$2.4m per day at the platform’s current volume. The percentage change in surplus does not depend on  $\beta_p$ , which enters the baseline and the counterfactual identically; only the conversion into dollars scales with it. The loss is larger when substitution within a capability tier is weaker, ranging from \$3.9m to \$1.3m per day as  $\sigma$  moves from 0 to 0.7 (Figure 4 and Table 5). It also depends on how large the outside option is taken to be, which is a modelling choice rather than a measurement: raising the outside share from the observed 2.4% to 50% raises the proportional surplus loss, from 21% to 47%, while lowering the dollar figure to \$1.0m per day. The price effect is invariant to both. The price effect is almost invariant to  $\sigma$ , because it is governed by the composition of the surviving choice set rather than by the strength of substitution. Concentration moves against the presumption that open weights fragment the market: the provider Herfindahl index rises from 897 to 1048 when open models are removed, because open-weight volume is spread across more suppliers than the proprietary volume that replaces it.

These numbers hold prices fixed. That is the right convention given the rigidity documented in Section 3.2, and it makes the exercise a statement about variety rather than about pricing conduct, but it understates the long-run effect. Allowing proprietary providers to reprice would require a pricing model, and the elasticity in Section 3.3 is not consistent with the static Bertrand model that would ordinarily supply one, so we leave that margin unpriced and read our estimates as a lower bound.

The left panel of Figure 4 repeats the exercise separately in each of the 29 task-level markets, where shares of spending are reported directly. The implied price increase rises almost one for one with the open-weight share of task spending, with a slope of 0.98. Averaged across tasks and weighted by spending, prices rise 18% and surplus falls 13%, with the largest effects in agentic and coding work, where open-weight models hold about a third of spending, and the smallest in data extraction and transformation, where they hold under a sixth. The task-level numbers are smaller than the aggregate because task shares are measured on spending, which weights the expensive proprietary models that open weights do not displace.

## 6 Discussion

The three facts fit together. Short-run demand for a given model is inelastic, so providers gain little by cutting the price of a model in the field and do not; because posted prices do not move, competition is expressed in the terms on which new models enter; and because the market is growing quickly, entrants can take a large share of volume without taking it from anyone. In that environment, the standard empirical signature of competitive discipline, a price response by incumbents, is absent



**Figure 4: The value of the open-weight choice set.** Left: for each of the 29 task markets, the price increase implied by removing open-weight models against their share of task spending; marker area is proportional to the task’s share of platform spending. Right: aggregate surplus loss per day across values of the within-tier substitution parameter and the price coefficient.

even though the constraint is economically large. Our counterfactual puts the value of the open-weight choice set at roughly one fifth of consumer surplus on this platform.

**A simple account.** The three fit together through the share of demand that is actually in play. Let  $M_t$  be total demand, growing at rate  $g$  per period, and let a fraction  $\phi$  of existing demand be revisited each period, whether because a workload is rebuilt or because a team re-evaluates its choice. Contestable demand is  $(g + \phi)M_t$ , and a model’s volume evolves as

$$y_{j,t+1} = (1 - \phi) y_{jt} + \sigma_j (g + \phi) M_t, \quad (4)$$

where  $\sigma_j$  is its share of contestable demand. Three implications follow. An incumbent loses volume only when  $\sigma_j(g + \phi)M_t < \phi y_{jt}$ , that is, when its share of contestable demand falls below  $\phi/(g + \phi)$  times its share of the stock; with  $g \approx 0.13$  per week in our window, an incumbent with an average share of new demand keeps growing. An entrant reaches a share of about  $\sigma_k(g + \phi)/(1 + g)$  after one period, so entrants can take a large share of volume quickly without anyone shrinking. And a permanent price change reaches only the contestable margin, so the response measured  $\tau$  periods out is attenuated by roughly  $1 - (1 - \phi - g)^\tau$  relative to its long-run value.

That last point bounds how much of the inelasticity is an artefact of slow adjustment. Taking  $\phi + g \approx 0.23$  per week and the six weekly observations over which Figure 2 shows the response settling, the attenuation factor is about 0.8, so the long-run elasticity implied by our estimate is near  $-0.6$ . Demand for a given model is inelastic even after adjustment. The static first-order condition of a provider maximising current profit is therefore not satisfied at the posted price, which leaves two readings: providers price against a margin we do not observe, such as capacity, hardware amortisation or the option value of usage; or the posted price is not chosen to maximise current profit at all. The rigidity in Section 3.2 favours the second.

This also reconciles two readings of the existing evidence. Nagle and Yue [2025] find that a great deal of proprietary usage sits on models dominated by an open alternative, and infer large foregone savings. That pattern is a puzzle if demand adjusts quickly; it is what one should expect if the short-run elasticity is about one-half and switching a workload is costly. The gains they identify are real, but they accrue as workloads turn over rather than as incumbents reprice.

**Limitations.** Our market is one router, and one that selects for price-sensitive developers, so the elasticities and the counterfactual describe the segment where substitution toward open weights is easiest. The capability index is built from third-party benchmarks that compress into a single dimension, and models that look identical on it are plainly not interchangeable in practice, which is precisely why the discrete-choice approach is needed and also why the index should not be read as a complete description of quality. We do not identify  $\beta_p$  and  $\sigma$  from the data and instead report sensitivity across a range; a design with exogenous variation in the choice set, such as a large release



299 observed over a longer window, would pin them down. The diversion test has power against moder-  
300 ate effects only. Finally, the counterfactual holds prices fixed, and is therefore silent about how the  
301 proprietary frontier would be priced in a world without open weights.

302 **Implications.** For policy, the result reframes what to measure. Debates about the competitive  
303 effect of open weights have proceeded on the presumption that the effect, if real, would show up in  
304 the prices proprietary providers charge. In a market where posted prices are set once and left alone,  
305 that presumption is wrong, and an assessment based on incumbent price responses will conclude  
306 that open weights do not matter. The relevant margin is the price at which the next unit of demand  
307 is served, and on that margin the constraint is substantial.

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## 333 A Data construction

334 **Endpoints.** The catalogue endpoint returns, for every listed model, the posted input and output  
335 price per token, context length, input and output modalities, a creation timestamp, and the Hugging  
336 Face repository identifier when the weights are public. The leaderboard endpoint returns tokens,  
337 requests, tool calls and media counts by model and serving variant. A benchmark endpoint returns  
338 third-party capability scores and usage-weighted effective prices. A task endpoint returns, for each  
339 of 29 usage categories, the category’s share of platform spending and the share held by each of its ten  
340 largest models, in both spending and token terms. An endpoint listing gives each hosting provider’s  
341 price, quantisation and uptime for a given model. All are public and unauthenticated.

342 **Daily usage.** Each model’s public page embeds its own trailing-month daily usage series in the  
343 page’s streamed render payload. We request one page per model with observed usage and parse the  
344 payload, which yields 457 models over 30 days. The same payload carries per-endpoint latency and  
345 throughput percentiles, which we aggregate to the model level by request count.

346 **Historical coverage.** Internet Archive captures supply history. Captures of the catalogue endpoint  
347 give a price and characteristic panel from July 2023. Captures of the leaderboard page embed the  
348 leaderboard payload in the same streamed format, so we recover model-level usage without relying  
349 on the rendered page. Two vintages are present and we treat them separately: captures before late  
350 2024 embed a series of non-overlapping weekly buckets, and later captures embed a single trailing-  
351 window snapshot. Shares are comparable across vintages; levels are not, so all specifications include  
352 date fixed effects, and Table 3 reports the two vintages separately. We fetch captures at a fixed  
353 spacing with adaptive backoff, which is the binding cost of the collection: the archive requests take  
354 a few hours, and every other step runs in minutes on a laptop.

355 **Merging.** Usage and price observations are matched by model identifier and nearest date within  
356 ten days. Openness is taken from the current catalogue where the model is still listed, because the  
357 weight-repository field is absent from the earliest captures. Prices are dollars per million tokens  
358 throughout. The fixed-weight price index used in (1) weights posted input and output prices by  
359 the sample-mean token mix, so that it carries no feedback from a model’s own usage composition;  
360 Table 3 shows the estimate is not sensitive to that choice.

## 361 **B Capability imputation**

362 Benchmark scores are missing for smaller models. For each missing dimension we fit a ridge regres-  
363 sion on the dimensions the model does have, using models scored on both, and predict the missing  
364 value. Masking half of the fully scored models and predicting the held-out values gives  $R^2$  of 0.99,  
365 0.97 and 0.96 for the reasoning, coding and agentic dimensions, with mean absolute errors of 1.7,  
366 5.0 and 3.1 points against standard deviations of 16.4, 22.6 and 18.5. Imputation raises coverage  
367 from 109 models and 77% of tokens to 180 models and 91% of tokens. All results in the main text  
368 hold on the directly scored subsample.

## 369 **C Robustness**

370 Table 3 reports the elasticity across price measures, outcome measures, capture vintages, openness  
371 and sample period. The estimate stays between  $-0.38$  and  $-0.63$ . The realised blended price, which  
372 mixes posted prices with the model’s own input-output composition, gives the largest magnitude, as  
373 expected if a shift toward prompt-heavy workloads lowers the average price per token and raises  
374 volume at the same time; the fixed-weight index removes that channel and is our preferred measure.

375 Two further checks bear on the elasticity. A price cut may accompany the launch of a replacement  
376 by the same provider, which would depress the old model’s volume for reasons other than price.  
377 Adding an indicator for observations within 28 days of a same-provider release, which covers 38%  
378 of the panel, leaves the estimate at  $-0.534$ ; dropping those observations gives  $-0.499$  (s.e. 0.123).

379 Table 4 reports the diversion test under three definitions of technological distance. None of the six es-  
380 timates is distinguishable from zero. The Euclidean and cosine measures give tighter standard errors  
381 than the Mahalanobis measure, and bound the differential decline for the most exposed incumbents  
382 at 0.13 and 0.09 log points respectively.

## 383 **D Counterfactual sensitivity**

384 Table 5 reports the counterfactual across the within-tier substitution parameter, and  
385 robustness. `json` reports it across the size of the outside option as well. The price effect is nearly  
386 invariant to  $\sigma$ , because it is driven by the composition of the surviving choice set rather than by  
387 the strength of substitution. The surplus loss falls as  $\sigma$  rises, because closer substitutes within a  
388 capability tier make the proprietary models better replacements for the open-weight models that  
389 leave.

**Table 3:** Own-price elasticity across specifications. Each row is a separate estimate of (1) with model and date fixed effects; standard errors clustered by model.

| Specification                 | Estimate | Std. error | Observations | Models |
|-------------------------------|----------|------------|--------------|--------|
| Fixed-weight price (baseline) | −0.534   | 0.102      | 19,803       | 634    |
| Input price                   | −0.526   | 0.096      | 19,803       | 634    |
| Output price                  | −0.485   | 0.109      | 19,803       | 634    |
| Realised blended price        | −0.658   | 0.095      | 19,803       | 634    |
| Requests rather than tokens   | −0.463   | 0.087      | 19,803       | 634    |
| Weekly captures only          | −0.472   | 0.123      | 15,133       | 432    |
| Trailing captures only        | −0.476   | 0.125      | 4,670        | 428    |
| Open-weight models            | −0.582   | 0.116      | 8,873        | 321    |
| Proprietary models            | −0.534   | 0.204      | 10,930       | 316    |
| 2025 onwards                  | −0.527   | 0.100      | 7,906        | 542    |

**Table 4:** Diversion at entry under alternative distance measures. Each row estimates (2) with incumbent-by-entrant and entrant-by-day fixed effects and post-release controls for the incumbent’s own capability and price.

| Distance    | Exposure                           | Estimate | Std. error | Observations |
|-------------|------------------------------------|----------|------------|--------------|
| Mahalanobis | similarity $\times$ relative value | 0.010    | 0.070      | 11,820       |
| Mahalanobis | similarity                         | 0.019    | 0.120      | 11,820       |
| Euclidean   | similarity $\times$ relative value | −0.006   | 0.047      | 11,820       |
| Euclidean   | similarity                         | 0.008    | 0.065      | 11,820       |
| Cosine      | similarity $\times$ relative value | 0.022    | 0.031      | 11,820       |
| Cosine      | similarity                         | 0.083    | 0.086      | 11,820       |

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**Table 5:** The open-weight counterfactual across substitution parameters, at  $\beta_p = -1$ . Prices are dollars per million tokens; the surplus loss is evaluated at the platform’s volume on the reference day.

| $\sigma$ | Price paid<br>(\$/M) | Price without open<br>(\$/M) | Surplus change<br>(%) | Surplus loss<br>(\$m per day) | HHI<br>(without open) |
|----------|----------------------|------------------------------|-----------------------|-------------------------------|-----------------------|
| 0.0      | 0.70                 | 1.82                         | −33.8                 | 3.9                           | 1109                  |
| 0.2      | 0.70                 | 1.81                         | −27.5                 | 3.2                           | 1082                  |
| 0.4      | 0.70                 | 1.80                         | −21.0                 | 2.4                           | 1048                  |
| 0.6      | 0.70                 | 1.79                         | −14.3                 | 1.7                           | 1006                  |
| 0.7      | 0.70                 | 1.78                         | −10.8                 | 1.3                           | 982                   |

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